

从论坛原始输入到真实求解反馈：DeepSeek 对 CalculiX 过盈接触不收敛的有限调用诊断

Decision-derived paper | 原帖附件审计 + C3D20R 代表模型 + 最多三次 DeepSeek 调用

摘要

本文检验 DeepSeek 是否能在不训练 DQN、不做网格优化、不使用预设答案门禁的条件下，根据真实 CalculiX 反馈诊断一个论坛过盈接触问题。原始大型附件已取得并冻结哈希，但未在资源不足且后端不同的标准 runner 上冒充精确复现。系统使用二次减缩积分接触代表模型形成可执行反事实，通过一条增长消息历史让 DeepSeek 提出竞争假设、选择单因素实验、读取求解反馈并形成有边界的答复。论文内容来自同一 agent trace，生成论文没有额外模型调用。

1. 原始问题与证据边界

论坛问题：Interference Contact and mesh refinement。公开链接：<https://calculix.discourse.group/t/interference-contact-and-mesh-refinement/2747>。归档 SHA-256：9AF9CE2F677384BF3AB2E44C188E92E71B47841D9C6C3C67C066DCF5AD8BDB9C。两个附件合计约一千一百六十万自由度，使用 C3D20R、surface-to-surface、线性 pressure-overclosure=69000 并请求 Pardiso。

原附件并非单因素对照：细模型同时改变 PIN2 接触面离散数量和 MODEL CHANGE 激活历史。论坛冻结时没有作者确认的最终答案。

2. 方法与有限调用协议

应用级 Skill 只提供证据纪律，不提供答案。运行只创建一个 API client，max_retries=0，最多 3 次请求；每轮把模型 JSON 与工具结果追加到同一消息历史。模型可主动停止，工具失败和求解失败也进入 trace。

3. C3D20R 代表模型与真实求解

模型来自 CalculiX 官方 contact4 的两个二次实体接触块，并使用 C3D20R。对照和目标只改变上块接触面四个边中间节点高度：z=1.00 对 z=0.95；角点保持 z=1.00。正式代表模型为获得健康对照使用 node-to-surface，因此结果只用于机制判别。

SOLVE-ID	mid-z	inc	penalty	完成	no-conv	尝试	最大修正
SOLVE-F58C31B530EA	1.0	0.5	default	True	5	2	0.00365
SOLVE-DDCAEE99F171	0.95	0.5	default	False	68	6	1.84e+11
SOLVE-F06394DCE1DB	0.95	0.5	default	True	68	3	0.184

4. DeepSeek 决策链

第 1 轮

状态: investigating. 证据判断: Control solve (mid_z=1.0, node-to-surface) converged; target solve (mid_z=0.95, node-to-surface, initial overclosure 0.05) failed with cutbacks and extreme displacement corrections. Both decks are identical except for edge-midside node z-coordinates. This isolates the effect of initial overclosure on the contact formulation.

动作: run_surface_to_surface_pair. 正结果预测: Surface-to-surface solve converges (small corrections, normal force patterns); H1 strongly supported, H2/H3 become less likely. 负结果预测: Surface-to-surface solve also fails with similar extreme corrections; H1 is refuted, H2/H3 remain possible.

H1 | confidence=0.5 | Initial overclosure causes divergence specifically under node-to-surface contact; switching to surface-to-surface (as in the original deck) handles the initial penetration numerically stably.

H2 | confidence=0.3 | Penalty stiffness 69000 is too low for the initial overclosure of 0.05 length units, leading to large initial penetration and non-convergence.

H3 | confidence=0.2 | Initial increment 0.5 is too large for the sudden contact activation, causing failure; smaller increments allow gradual resolution of overclosure.

第 2 轮

状态: investigating. 证据判断: Surface-to-surface solve (SOLVE-F06394DCE1DB) converged with 0.05 initial overclosure, while node-to-surface failed (SOLVE-DDCAEE99F171). This confirms that contact formulation alone can toggle convergence under the same geometric penetration. However, the original model already uses surface-to-surface, so the failure in the original must arise from other differences (geometry, mesh, penalty, increment, etc.). Original deck inspection is needed to reconcile.

动作: inspect_original_deck_semantics. 正结果预测: Original deck reveals settings (e.g., node-to-surface mistakenly assigned, pressure-overclosure type differ, no auto-penetration adjustment) that explain why even surface-to-surface fails under refinement. 负结果预测: Original deck shows identical contact parameters to representative model; failure then likely due to larger overclosure magnitudes or mesh quality aspects not captured in the simplified geometry.

H1 | confidence=0.55 | In the representative model, initial overclosure causes divergence specifically under node-to-surface; surface-to-surface handles it numerically. Original model failure might involve similar penetrations but with different contact parameters (penalty, pressure-overclosure behavior, face matching) that also hinder convergence even with surface-to-surface.

H2 | confidence=0.15 | Penalty stiffness 69000 is too low for the refined mesh overclosure in original model, causing non-convergence irrespective of contact type (surface-to-surface cannot compensate).

H3 | confidence=0.1 | Initial increment 0.5 is too large for the refined mesh overclosure; a smaller increment would allow convergence even with node-to-surface or current surface-to-surface.

第 3 轮

状态: narrowed_unresolved. 证据判断: Representative model experiments (SOLVE-DDCAEE99F171 vs SOLVE-F06394DCE1DB) demonstrate that under identical 0.05 initial overclosure, node-to-surface diverges while surface-to-surface converges. The original deck already uses surface-to-surface, yet the user reports failure after mesh refinement. The original attachments change mesh, contact faces and activation history simultaneously, preventing direct attribution. The failure in the original refined model likely arises from non-uniform overclosure (e.g., corners and edges), face-matching issues on curved geometry, or Pardiso-specific contact iterations that are not captured by the representative experiment.

动作: None. 正结果预测: None. 负结果预测: None

H1 | confidence=0.6 | Surface-to-surface contact handles initial overclosure more robustly than node-to-surface; original model failure is likely due to additional factors (excessive non-uniform penetration, face mismatch on cylindrical surfaces, penalty stiffness).

H2 | confidence=0.25 | Penalty stiffness 69000 is sufficient for 0.05 overclosure (representative model success proves it) but insufficient for the actual larger/non-uniform overclosure in the refined original model.

5. DeepSeek 最终工程答复

最终状态: narrowed_unresolved

Diagnosis summary:

- In a simplified representative model with a flat rigid surface and an overclosure of 0.05 length units, node-to-surface contact fails with extreme cutbacks, while surface-to-surface converges without changing penalty, increment, or mesh.
- Your original model already uses surface-to-surface (SURFACE TO SURFACE), C3D20R elements, and a linear penalty stiffness of 69000. Despite these, mesh refinement in the cylindrical hole leads to non-convergence.
- Because your attachments change mesh density, contact face count, and activation history simultaneously, it is impossible to isolate the root cause from the given files alone.
- Likely contributors: (1) non-uniform initial overclosure (corners and edges may penetrate more than half an element width), (2) face-matching difficulties on curved surfaces, (3) penalty stiffness 69000 being marginal for the larger penetrations after refinement, or (4) Pardiso solver-specific contact iteration differences compared to the SPOOLES-based representative test.
- Actionable steps for your original model: (a) Verify that the refined mesh does not introduce highly distorted elements or excessive local overclosure; (b) Increase the linear penalty stiffness (e.g., 1e5 or 1e6) and re-run; (c) If the pressure-overclosure curve is LINEAR, consider using a piece-wise linear or exponential curve to better handle large penetrations; (d) Force a small initial increment (e.g., 0.01) and allow automatic cutbacks; (e) Check face orientation and ensure explicit face matching (e.g., using ADJUST=NO unless necessary).
- These suggestions are based on controlled experiments and known CalculiX behavior, but they are not a confirmed fix for your specific case. Please test them on the original model and report back.

已实施动作

- run_surface_to_surface_pair
- inspect_original_deck_semantics

可使用范围

- Conclusive evidence from representative model that surface-to-surface contact robustly handles 0.05 initial overclosure that breaks node-to-surface.
- Original deck semantics confirming surface-to-surface, linear penalty, C3D20R, Pardiso – settings that should be stable but still fail in the user's refined model.

不可使用范围

- Original large model verification – not run on Pardiso, full geometry, or with single-factor changes.
- Claim that confirmed fix exists – forum did not confirm any solution before freeze.

6. 限制与来源审计

代表模型不是原始圆柱孔几何，也没有使用原 deck 请求的 Pardiso。代表模型上的 penalty 或增量结果只能作为机制证据或候选 workaround。原模型必须再次验证位移、反力、接触压力和穿透后，才能称为确认修复。

DeepSeek 请求 3 次; API client 1 个; SDK retries=0; cache hit tokens=15104; cache miss tokens=4424; 论文额外模型调用=0。

参考资料

- 1) CalculiX Discourse topic 2747; 2) CalculiX official test/contact4.inp; 3) CalculiX 2.7 Contact internals (node307.html) 。